

**The University of Texas at Austin**  
**IN-CLASS WORK 7**

*Introduction to Financial Mathematics*

September 23, 2026

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**EUROPEAN PUT OPTIONS.**

**Problem 7.1.** (2 points) In which of the following option positions is the investor exposed to an unlimited loss?

- a. Long put option.
- b. Short put option.
- c. Long call option.
- d. Short call option.
- e. None of the above.

**Problem 7.2.** The initial price of the market index is \$900. After 3 months the market index is priced at \$915. The nominal rate of interest convertible monthly is 4.8%. The premium on the put, with a strike price of \$930, is \$8.00. What is the profit at expiration for a **long** put?

- a. \$15.00 loss.
- b. \$6.90 loss.
- c. \$6.90 gain.
- d. \$15.00 gain.
- e. None of the above.

**Problem 7.3.** (3 points) The initial price of the market index is \$1000. After 3 months the market index is priced at \$950. The nominal rate of interest convertible quarterly is 4.0%.

The premium on the long put, with a strike price of \$975, is \$10.00. What is the profit at expiration for this long put?

- a. \$12.00 loss
- b. \$14.90 loss
- c. \$12.00 gain
- d. \$14.90 gain
- e. None of the above.

**Problem 7.4.** *Sample FM(DM) #12*

Consider a European put option on a stock index without dividends, with 6 months to expiration, and a strike price of 1,000. Suppose that the nominal annual risk-free rate is 4% **convertible semiannually**, and that the put costs 74.20 now. What price must the index be in 6 months so that being long the put would produce the same profit as being short the put?

- a. \$922.83.
- b. \$924.32.
- c. \$1,000.00.
- d. \$1,075.68.
- e. \$1,077.17.

**Problem 7.5.** (3 points) *Source: Sample FM(DM) Problem #62.*

The stock price today equals \$100 and its price in one year is modeled by the following distribution:

$$S(1) \sim \begin{cases} 125 & \text{with probability } 1/2 \\ 60 & \text{with probability } 1/2 \end{cases}$$

The annual effective interest rate equals 3%.

Consider an at-the-money, one-year European put option on the above stock whose initial premium is equal to \$7.

What is the expected profit of this put option?